

Lab 6

Question 1

How does Joey compare to Sockeye on the WMT newstest2017 English to German benchmark in terms of BLEU score? How does a Joey Transformer compare with the best Joey RNN setting?

In english to german benchmark in terms of bleu score, Joey beats Sockeye in Groundhog RNN by 0.3 and in Best RNN by 0.4. For Transformer, Sockeye beats Joey by 0.1. The Joey Transformer is better than the best Joey RNN by 1.4 BLEU score.

Describe the user study. How did "experts" compare with "novices" in the user study?

The user study was to see how easy it was for people to learn using the tool. The user study took people with different levels of machine translation experience to conduct the user study. They used two types of people, "experts" and "novices". Experts are people with experience with machine translation or deep learning frameworks. Novices are people with no experience with machine translation tools.

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Question 2

```
!pip install EasyNMT
```

```
import nltk
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
True
```

```
from easynmt import EasyNMT
```

```
opusModel = EasyNMT('opus-mt')
```

```
from google.colab import drive
gdrive_mount = '/content/drive'
drive.mount(gdrive_mount, force_remount=True)
```

```
Mounted at /content/drive
```

```
labdir = '/content/drive/My Drive/Teaching/605.646/Fall 2025/lab06/spanish'
%cd "$labdir"
```

```
/content/drive/My Drive/Teaching/605.646/Fall 2025/lab06/spanish
```

```
with open('flores101.test.es.txt', 'r', encoding='utf-8') as f:
    spanishSentences=[line.strip() for line in f if line.strip()]
```

```
opusTranslations = opusModel.translate(spanishSentences, source_lang='es', target_lang='en')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
tokenizer_config.json: 100% 44.0/44.0 [00:00<00:00, 4.69kB/s]
source.spm: 100% 826k/826k [00:00<00:00, 48.2MB/s]
target.spm: 100% 802k/802k [00:00<00:00, 42.9MB/s]
vocab.json: 1.59M/? [00:00<00:00, 60.7MB/s]
config.json: 1.44k/? [00:00<00:00, 150kB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:175: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: 100% 312M/312M [00:05<00:00, 50.2MB/s]
generation_config.json: 100% 293/293 [00:00<00:00, 35.6kB/s]
model.safetensors: 100% 312M/312M [00:04<00:00, 44.8MB/s]
```

```
for i in range(10):
    print(f"Opus Translation {i+1}:")
    print("Spanish - " + spanishSentences[i])
    print("English - "+opusTranslations[i])
    print("\n")
```

Opus Translation 1:
Spanish - «Actualmente, tenemos ratones de cuatro meses de edad que antes solían ser diabéticos y que ya no lo son», agregó.
English - "Currently, we have four-month-old mice who used to be diabetic and who are no longer diabetic," he added.

Opus Translation 2:
Spanish - La investigación todavía se ubica en su etapa inicial, conforme indicara el Dr. Ehud Ur, docente en la carrera de medicina de la Universidad de Dalhousie, en Halifax, Nueva Escocia.
English - The research is still in its early stages, as indicated by Dr. Ehud Ur, professor of medicine at Dalhousie University, Halifax, Nova Scotia, and director of the clinical research program.

Opus Translation 3:
Spanish - Al igual que otros especialistas, es escéptico acerca de si la diabetes tiene cura y remarca que estos descubrimientos no son relevantes para quienes que ya padecen de diabetes tipo 1.
English - Like other specialists, she is skeptical about whether diabetes has a cure and points out that these findings are not relevant to those who already have type 1 diabetes.

Opus Translation 4:
Spanish - El lunes, Sara Danius, secretaria permanente del Comité por el Nobel de Literatura de la Academia Sueca, declaró públicamente en un programa radiofónico de Sveriges, en Suecia.
English - On Monday, Sara Danius, permanent secretary of the Nobel Committee for Literature at the Swedish Academy, publicly stated in a Sveriges radio program in Sweden that, by the end of the year, the research program will be completed.

Spanish - Danius declaró: «Actualmente no estamos haciendo nada. Llamé y envié correos electrónicos a su asistente más cercano y recibí contestaciones muy cordiales. Por el momento, eso es ciertamente suficiente».

English - Danius stated: "We are currently doing nothing. I called and e-mailed his closest assistant and received very cordial responses. For the moment, that is certainly enough."

English - Previously, Ring's executive director, Jamie Siminoff, mentioned that the company had started because the doorbell on his entrance couldn't be heard from the store he had.

Spanish - Él dijo que construyó un timbre wifi.
English - He said he built a Wi-Fi bell.

English - Siminoff stated that sales skyrocketed after his appearance in an episode of Shark Tank in 2013, in which the TV show panel refused to fund the company's start-up.

English - Towards the end of 2017, Siminoff made an appearance on the TV shopping channel QVC.

English - Ring also reached an agreement in the dispute with the competition security company, ADT Corporation.

```
M2MTranslations = M2Mmodel.translate(spanishSentences, source lang='es', target lang='en')
```

```
for i in range(10):
    print(f"M2M Translation {i+1}:")
    print("Spanish - " + spanishSentences[i])
    print("English - " + M2MTranslations[i])
    print("\n")
```

Spanish - "Actualmente, tenemos ratones de cuatro meses de edad que antes solían ser diabéticos y que ya no lo son", agregó.
English - "At present, we have four-month-old mice that previously used to be diabetic and that they are no longer," he added.

Spanish - La investigación todavía se ubica en su etapa inicial, conforme indicara el Dr. Ehud Ur, docente en la carrera de medicina de la Universidad de Dalhousie, en Halifax, Nueva Escocia, y director de la investigación.

Spanish – Al igual que otros especialistas, es escéptico acerca de si la diabetes tiene cura y remarca que estos descubrimientos no son relevantes para quienes ya padecen de diabetes.
English – Like other specialists, he is skeptical about whether diabetes has a cure and notes that these discoveries are not relevant for those who already suffer from type 1 diabetes.

Spanish – El lunes, Sara Danius, secretaria permanente del Comité por el Nobel de Literatura de la Academia Sueca, declaró públicamente en un programa radiofónico de Sveriges, en English – On Monday, Sara Danius, the Permanent Secretary of the Swedish Academy's Nobel Prize in Literature Committee, publicly declared in a Sveriges radio program, that, by not

English - Danius said: "At the moment we're not doing anything. I called and sent emails to your closest assistant and received very cordial answers. For now, that is undoubtedly

Spanish - Con anterioridad, el director ejecutivo de Ring, Jamie Siminoff, mencionó que la compañía había iniciado a raíz de que el timbre de su entrada no podía oírse desde la t...

Spanish - Él dijo que construyó un timbre wifi.
English - He said he built a wifi timbre.

English - Siminoff said the sales dropped after it appeared in a Shark Tank episode in 2013, in which the TV panel refused to fund the company's launch.

Spanish - Hacia fines del año 2017, Siminoff hizo una aparición en el canal de compras televisadas QVC.
English - By the end of 2017, Siminoff made a appearance on the QVC television shopping channel.

English - Ring also reached an agreement in the dispute with the competition security company, ADT Corporation.

```
!pip install sacrebleu
```

```
Collecting sacrebleu
  Downloading sacrebleu-2.5.1-py3-none-any.whl.metadata (51 kB)
    _____ 51.8/51.8 kB 5.0 MB/s eta 0:00:00
Collecting portalocker (from sacrebleu)
  Downloading portalocker-3.2.0-py3-none-any.whl.metadata (8.7 kB)
Requirement already satisfied: regex in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (2024.11.6)
Requirement already satisfied: tabulate>=0.8.9 in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (0.9.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (2.0.2)
Collecting colorama (from sacrebleu)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (5.4.0)
Downloading sacrebleu-2.5.1-py3-none-any.whl (104 kB)
    _____ 104.1/104.1 kB 12.3 MB/s eta 0:00:00
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading portalocker-3.2.0-py3-none-any.whl (22 kB)
Installing collected packages: portalocker, colorama, sacrebleu
Successfully installed colorama-0.4.6 portalocker-3.2.0 sacrebleu-2.5.1
```

```
with open('flores101.test.en.txt', 'r', encoding='utf-8') as f:
    englishSentences=[line.strip() for line in f if line.strip()]

opusTranslationsLower = [s.lower() for s in opusTranslations]

M2MTranslationsLower = [s.lower() for s in M2MTranslations]

englishSentencesLower = [s.lower() for s in englishSentences]

import sacrebleu

opusBLEU = sacrebleu.corpus_bleu(opusTranslationsLower,[englishSentencesLower])

print(f"Opus BLEU score: {opusBLEU.score}")

Opus BLEU score: 28.55677582670647

opusChrF= sacrebleu.corpus_chrf(opusTranslationsLower,[englishSentencesLower])

print(f"Opus ChrF score: {opusChrF.score}")

Opus ChrF score: 59.0562709544656

opusTER = sacrebleu.corpus_ter(opusTranslationsLower,[englishSentencesLower])

print(f"Opus TER score: {opusTER.score}")

Opus TER score: 60.905894708004205

M2MBLEU = sacrebleu.corpus_bleu(M2MTranslationsLower,[englishSentencesLower])

print(f"M2M BLEU score: {M2MBLEU.score}")

M2M BLEU score: 23.765155312240157

M2MChrF = sacrebleu.corpus_chrf(M2MTranslationsLower,[englishSentencesLower])

print(f"M2M ChrF score: {M2MChrF.score}")

M2M ChrF score: 55.759612965182285

M2MTER = sacrebleu.corpus_ter(M2MTranslationsLower,[englishSentencesLower])

print(f"M2M TER score: {M2MTER.score}")

M2M TER score: 66.39879457559016
```

Question 4

sentences = ["Berlin straddles the banks of the River Spree, which flows into the River Havel (a tributary of the River Elbe) in the western borough of Spandau.", "Among the city's main topographical features are the many lakes in the western and southeastern boroughs formed by the Spree, Havel, and Dahme rivers.", "Due to its location in the European Plain, Berlin is influenced by a temperate seasonal climate."]

```
'or i in range(len(sentences)):
    print(f"Sentence {i+1}:")
    print(f"Sentence: {sentences[i]}")
    spanishSentence = opusModel.translate(sentences[i],source_lang='en', target_lang='es')
    print(f"Spanish: {spanishSentence}")
    frenchSentence = opusModel.translate(spanishSentence,source_lang='es', target_lang='fr')
    print(f"French: {frenchSentence}")
    russianSentence = opusModel.translate(frenchSentence,source_lang='fr',target_lang='ru')
    print(f"Russian: {russianSentence}")
    englishSentence = opusModel.translate(russianSentence,source_lang='ru',target_lang='en')
    print(f"Final English: {englishSentence}")
    print("\n")
```

Sentence 1:
Sentence: Berlin straddles the banks of the River Spree, which flows into the River Havel (a tributary of the River Elbe) in the western borough of Spandau.
Spanish: Berlín se extiende por las orillas del río Spree, que desemboca en el río Havel (afluente del río Elba) en el distrito occidental de Spandau.
French: Berlin s'étend sur les rives de la rivière Spree, qui débouche sur la rivière Havel (affluent de l'Elbe) dans le district occidental de Spandau.
Russian: Берлин распространяется на берега реки Спре, которая приводит к реке Хэвел в западной части Спандоу.
Final English: Berlin extends to the banks of the Spree River, which leads to the Havel River in the western Spandow.

Sentence 2:
Sentence: Among the city's main topographical features are the many lakes in the western and southeastern boroughs formed by the Spree, Havel, and Dahme rivers.
Spanish: Entre las principales características topográficas de la ciudad se encuentran los muchos lagos en los distritos del oeste y sureste formados por los ríos Spree, Havel y Dahme.
French: Parmi les principales caractéristiques topographiques de la ville figurent les nombreux lacs dans les districts de l'ouest et du sud-est formés par les rivières Spree, Havel et Dahme.
Russian: К числу основных топографических характеристик города относятся многочисленные озера в западных и юго-восточных районах, образованные реками Спре, Хавел и Дахме.
Final English: The main topography characteristics of the city include numerous lakes in the western and south-eastern regions, formed by the Rivers Spree, Havel and Dahma.

Sentence 3:
Sentence: Due to its location in the European Plain, Berlin is influenced by a temperate seasonal climate.
Spanish: Debido a su ubicación en la llanura europea, Berlín está influenciada por un clima estacional templado.
French: En raison de son emplacement dans la plaine européenne, Berlin est influencée par un climat saisonnier tempéré.
Russian: Из-за того, что Берлин находится на европейской равнине, он страдает умеренным сезонным климатом.
Final English: Because Berlin is on the European plain, it suffers from a moderate seasonal climate.

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From the above translated statements, we can see that the meaning of the sentences are preserved for the most part. In the first sentence, the translated sentence cuts out the phrase in parentheses which says "(a tributary of the River Elbe)". There are also spelling mistakes in the names of things, for example, in sentence 1, "Spandau" is made into "Spandow". In sentence 2, "Dahme" river is made into "Dahma" river.

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